Network error scaling with number of units

SLANT network averaging note, 31 October 2026. Carthage College Space Sciences.

The claim

The proposal's Projected cost and scalability section states: "If at most 1% of each unit's offset variance is shared, the hundred-unit mean carries 0.32% between-unit error against 2.3% for one unit." This note gives the model behind that sentence, the inputs and their sources, the arithmetic, and what the figure leaves out. Page 5, panel (c) of the visual PDF plots the same model.

The model

Each unit's mass density at a given altitude differs from the truth by two errors.

1. A random error e in each occultation, independent between occultations, with standard deviation R.
2. A repeatable offset b that stays with the unit, with standard deviation U across units. Calibration and detector drift set it.

The offset of unit k is split into a part c that every unit shares and a part d_k of its own:

$b_k = c + d_k$, with $\mathrm{Var}(c) = f\,U^2$ and $\mathrm{Var}(d_k) = (1 - f)\,U^2$.

The fraction f is the share of the offset variance common to all units. Averaging n occultations spread evenly over N units leaves the shared part c unchanged and divides the rest:

$\mathrm{Var}(\text{mean}) = R^2/n + U^2\,[\,f + (1 - f)/N\,]$.

As n grows the random term vanishes, and the between-unit error of the network mean approaches $U\,\sqrt{(\,f + (1 - f)/N\,)}$. However many units fly, it cannot fall below the shared floor $U\,\sqrt{f}$.

Inputs

Input	Value	Source
R, random error per occultation	8.7 %	Mass density at 200 km in a severe storm (ap 400), one unit, one occultation: Technology performance section and Figure 2.
U, repeatable offset of one unit	2.3 %	The network study of 8 October (Network coverage and cross-calibration, on the Cost and scalability tab) gives 2.25 % for the repeatable offset of a single unit in a severe storm, for retrieved atomic oxygen at 200 km. The note carries the same value for mass density; no separate computation for mass density has been made.
f, shared fraction of the offset variance	at most 0.01	An assumption the project adopts. At f = 0.01 the shared part is at most one tenth of the offset in amplitude. No unit has flown, so f cannot yet be measured.
Occultations per unit per day	10.33	Annual average for a wake-facing unit with the slit plate's field of view: Occultation rate study, on the Cost and scalability tab.

Results

Between-unit error of the network mean, mass density at 200 km, severe storm.

Units, *N*	Long average, *f* = 0.01	One day, *f* = 0.01	Long average, *f* = 0	Long average, *f* = 0.1
1	2.30 %	3.55 %	2.30 %	2.30 %
6	0.96 %	1.47 %	0.94 %	1.15 %
30	0.48 %	0.69 %	0.42 %	0.83 %
100	**0.32 %**	0.42 %	0.23 %	0.76 %
500	0.25 %	0.28 %	0.10 %	0.73 %
Shared floor	0.23 %		0	0.73 %

"Long average" lets *n* grow without limit, so only the offset terms remain. "One day" uses *n* = 10.33 *N*, one day of occultations from every unit.

At 100 units: 2.3 % × √(0.01 + 0.99/100) = 2.3 % × √0.0199 = 0.32 %. The shared floor is 2.3 % × √0.01 = 0.23 %, and the independent part is 2.3 % × √(0.99/100) = 0.23 %. The two add in quadrature to 0.32 %.

The last two columns show how much rests on the assumed *f*. With no shared error, 100 units reach 0.23 %; with ten times the assumed share, they reach 0.76 %, almost all of it the 0.73 % shared floor.

Why a small shared fraction is plausible

1. The largest per-unit term is the drift of each detector's flat field, which belongs to that detector alone.
2. Filter films and detectors can be drawn from several production lots, which turns an error common to one lot into an error that differs between units.
3. Coincident occultations from two units measure the difference between their offsets (Other considerations section), so in flight the independent part of each offset shrinks further. That step reduces *U* and is outside this note.

Shared sources not yet estimated

Two sources could raise *f* above 0.01 and have no estimate yet: the NIST SURF III calibration shared by each 20-unit lot, and a uniform error in the N_2 absorption cross section.

What the figure leaves out

1. **Retrieval bias.** Errors that every unit makes identically through the retrieval, such as the bias from the prior that Figure 2 reports and cross-section errors, do not average down. They add to the values in this note.
2. **Physical variability.** The model averages measurements of one common density. A real network samples a changing atmosphere at different places and times.

3. **Even sampling and steady offsets.** The model assumes occultations spread evenly over the units and offsets that hold steady over the averaging time.

To reproduce

```
import math
R, U, f = 8.7, 2.3, 0.01          # per cent; f is the shared fraction of offset variance
def network_error(N, n=None):
    offset = U**2 * (f + (1 - f) / N)
    return math.sqrt(offset + (R**2 / n if n else 0.0))
print(round(network_error(100), 2))              # 0.32, long average
print(round(network_error(100, 10.33 * 100), 2)) # 0.42, one day
```